

113. If $x = \frac{a}{a-1}$ and $y = \frac{1}{a-1}$, then

- (a) x is equal to y
- (b) x is equal to y only if $a < 1$
- (c) x is greater than y
- (d) x is greater than y only if $a < 1$
- (e) y is greater than x only if $a < 1$

114. If $0 < a < 1$, then the value of $a + \frac{1}{a}$ is

- (a) less than 2
- (b) greater than 2
- (c) less than 4
- (d) greater than 4

115. If $\frac{a}{x} + \frac{y}{b} = 1$ and $\frac{b}{y} + \frac{z}{c} = 1$, then $\frac{x}{a} + \frac{c}{z}$ will be equal to:

- (a) 0
- (b) $\frac{b}{y}$
- (c) 1
- (d) $\frac{y}{b}$

116. If a, b, c are integers; $a^2 + b^2 = 45$ and $b^2 + c^2 = 40$, then the values of a, b and c respectively are

- (a) 2, 6, 3
- (b) 3, 2, 6
- (c) 5, 4, 3
- (d) None of these

117. If $x + y = 15$ and $xy = 56$, then what is the value of $x^2 + y^2$? (L.I.C.A.D.O., 2007)

- (a) 110
- (b) 113
- (c) 121
- (d) Cannot be determined
- (e) None of these

118. If $(a - b) = 4$ and $ab = 2$, then $(a^2 + b^2) = ?$

- (a) 18
- (b) 20
- (c) 25
- (d) None of these

119. If $a = bc$ and $c = a - b$, then the value of a is

(R.R.B., 2008)

- (a) $b^2 - 1$
- (b) $\frac{b^2}{b-1}$
- (c) $\frac{b}{b-1}$
- (d) None of these

120. If $a + b + c = 0$, find the value of

$$\frac{a^2}{(a^2 - bc)} + \frac{b^2}{(b^2 - ca)} + \frac{c^2}{(c^2 - ab)}.$$

(M.A.T., 2005)

- (a) 0
- (b) 1
- (c) 2
- (d) 4

121. If $\frac{x}{y} = \frac{a+2}{a-2}$, then $\frac{x^2 - y^2}{x^2 + y^2}$ is equal to

- (a) $\frac{8a}{a^2 + 4}$
- (b) $\frac{4a}{a^2 - 4}$
- (c) $\frac{4}{a^2}$
- (d) $\frac{4a}{a^2 + 4}$

122. If $3x + 7 = x^2 + P = 7x + 5$, what is the value of P ?

- (a) $\frac{1}{2}$
- (b) $8\frac{1}{4}$
- (c) $8\frac{1}{2}$
- (d) Cannot be determined

123. If $\frac{2a+b}{a+4b} = 3$, then find the value of $\frac{a+b}{a+2b}$.

- (a) $\frac{2}{7}$
- (b) $\frac{5}{9}$
- (c) $\frac{10}{7}$
- (d) $\frac{10}{9}$

124. If $(2a + 3b)(2c - 3d) = (2a - 3b)(2c + 3d)$, then

- (a) $\frac{a}{b} = \frac{c}{d}$
- (b) $\frac{a}{d} = \frac{c}{b}$
- (c) $\frac{a}{b} = \frac{d}{c}$
- (d) $\frac{b}{a} = \frac{c}{d}$

125. If $(a + b + 2c + 3d)(a - b - 2c + 3d) = (a - b + 2c - 3d)(a + b - 2c - 3d)$, then $2bc$ is equal to

- (a) $\frac{3}{2}$
- (b) $\frac{3a}{2d}$
- (c) $3ad$
- (d) a^2d^2

126. The expression $\frac{x+y}{x-y} \div \frac{(x+y)^2}{(x^2 - y^2)}$ is equal to

- (a) -1
- (b) $x - y$
- (c) 1
- (d) $x + y$

127. $\frac{a^2 - b^2 - 2bc - c^2}{a^2 + b^2 + 2ab - c^2}$ is equivalent to

- (a) $\frac{a-b+c}{a+b+c}$
- (b) $\frac{a-b-c}{a-b+c}$
- (c) $\frac{a-b-c}{a+b-c}$
- (d) $\frac{a+b+c}{a-b+c}$

128. If $x + y + z = 0$, then $x^2 + xy + y^2$ equals

- (a) $y^2 + yz + z^2$
- (b) $y^2 - yz + z^2$
- (c) $z^2 - zx + x^2$
- (d) $z^2 + zx + x^2$

129. If $a + b + c = 2s$, then the value of $(s - a)^2 + (s - b)^2 + (s - c)^2 + s^2$ will be

- (a) $s^2 - a^2 - b^2 - c^2$
- (b) $s^2 + a^2 + b^2 + c^2$
- (c) $a^2 + b^2 + c^2$
- (d) $4s^2 - a^2 - b^2 - c^2$

130. If $(x + y)^2 - z^2 = 4$, $(y + z)^2 - x^2 = 9$, $(z + x)^2 - y^2 = 36$, what is/are the value(s) of $x + y + z$?

- (a) 0
- (b) ± 1
- (c) ± 3
- (d) ± 7